PHY101: Introduction to Physics-1

Monsoon, 2018

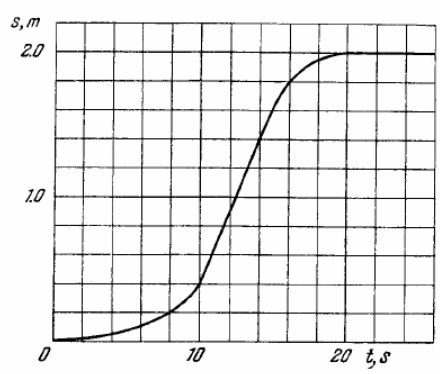
Tutorial-1

1. The position x of a particle undergoing simple harmonic motion is given as a function of time t as,

$$x = a \sin(\omega t + \varphi)$$

Using dimensional analysis figure out the dimensions of a, ω and φ.

2.The distance (s) travelled by an object is related with time (t) as, $s = 2t - 3t^2 + 4t^3$ where s is in metre and t is in second. What would be the velocity and acceleration after 2 seconds of its motion starting from the rest?

3. A particle in motion has its velocity v at a given time t. If the distance (s) travelled by the particle is expresses as, $s = \frac{1}{2}\, vt$. Show that the particle is moving with a constant acceleration.

4. A point moves rectilinearly in one direction. In the graph, the distance s traversed by the point as a function of time t is shown. Using the plot find the average velocity of the point during the time of motion. Also, indicate the region of maximum velocity in the graph and calculate the value.